# Structure and function of carbohydrates: (DLA)

1. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. A routine urinalysis showed high levels of a reducing sugar in urine that is not glucose. Which of the following best represents the chemical nature of the urine sugar present in this patient?
   A. A ketohexose
   B. An aldohexose
   C. A sugar acid
   D. A sugar alcohol
   E. An amino sugar

2. A 2-day-old boy is brought to the physician by his parents because of a history of diarrhea since birth. Clinical and laboratory studies show inherited deficiency of intestinal lactase. He is advised the avoidance of a dietary carbohydrate. Which of the following best represents the chemical nature of the carbohydrate that is most likely avoided in this patient?
   A. A non-reducing disaccharide
   B. Made up of two glucose units
   C. Made up of glucose and fructose
   D. Made up of glucose and galactose
   E. Monosaccharides are linked by alpha 1-4 linkage

3. A researcher is analyzing the dietary components of a 35-year-old man. Analysis shows a dietary component that increases the bulk of the feces and is able to adsorb water. Which of the following best describes the chemical nature of this dietary component?
   A. Has glucose and fructose units
   B. Has alpha 1-4 and alpha 1-6 glycosidic linkages
   C. Branched polysaccharide
   D. Has beta 1-4 glycosidic links
   E. A storage polysaccharide found in liver

4. A 2-year-old boy is brought to the physician by his mother because of a 3-week history of tripping over objects. Physical examination shows bilateral cataracts. Laboratory tests confirm inherited galactokinase deficiency. Examination of the lens shows an abnormal accumulation of a metabolite. Which of the following enzymes is most likely required for the formation of this metabolite in the lens of this patient?
   A. Glucose 6-phosphatase
   B. Hexokinase
   C. Glucose 6-phosphate dehydrogenase
   D. Aldose reductase
   E. Pyruvate dehydrogenase

5. A 62-year-old man comes to the physician because of a 1-month history of poor vision especially at night. He has a 15-year history of type 2 diabetes mellitus. Physical examination shows bilateral opacity of the lens. Laboratory studies show HbA1c is 9%. Which of the following best represents the chemical nature of the compound most likely responsible for the cataracts in this patient?
   A. A sugar acid
   B. An amino sugar
   C. A homopolysaccharide
   D. A heteropolysaccharide
   E. A sugar alcohol

Answer Key:

1. B
2. D
3. D
4. D
5. E

## Short explanations:

1. A 5-day-old girl is brought to the physician by her parents for a follow-up examination. A routine urinalysis showed high levels of a reducing sugar in urine that is not glucose. Which of the following best represents the chemical nature of the urine sugar present in this patient? Explanation – this infant most likely has galactosemia with presence of galactose in the urine
   A. A ketohexose – fructose is not likely as a 5-day-old infant's diet does not contain fructose
   **B. An aldohexose – CORRECT ANSWER – Galactose is a reducing sugar**
   C. A sugar acid – example uronic acid is a component of glycosaminoglycans
   D. A sugar alcohol – Galactitol is found in lens in children with galactosemia
   E. An amino sugar – example galactosamine or glucosamine are components of glycosaminoglycans

2. A 2-day-old boy is brought to the physician by his parents because of a history of diarrhea since birth. Clinical and laboratory studies show inherited deficiency of intestinal lactase. He is advised the avoidance of a dietary carbohydrate. Which of the following best represents the chemical nature of the carbohydrate that is most likely avoided in this patient?

Comment: The patient has congenital lactose intolerance. Management is avoidance of dietary lactose and stop breast feeding.

A. A non-reducing disaccharide – Sucrose is an example. Sucrose avoidance is advised in patients with fructose metabolism disorders
B. Made up of two glucose units – Maltose has two glucose units
C. Made up of glucose and fructose - Sucrose is an example. Sucrose avoidance is advised in patients with fructose metabolism disorders
**D. Made up of glucose and galactose – CORRECT ANSWER. Lactose is also a reducing sugar. Linkage is beta 1→ 4. Lactase (disaccharidase) hydrolyzed beta 1→4 links in lactose.**
E. Monosaccharides are linked by alpha 1-4 linkage – This linkage is found in maltose

3. A researcher is analyzing the dietary components of a 35-year-old man. Analysis shows a dietary component that increases the bulk of the feces and is able to adsorb water. Which of the following best describes the chemical nature of this dietary component?

Comment: The question is testing on chemical nature of cellulose which is aka dietary fiber. Cellulose in an unbranched homopolysaccharide. The glucose units are linked by beta 1→ 4 glycosidic linkages. Cellulose is found only in plant sources and not in animals.

A. Has glucose and fructose units – Cellulose is a homopolysaccharide of glucose
B. Has alpha 1-4 and alpha 1-6 glycosidic linkages – Starch and glycogen have alpha 1-4 and alpha 1-6 linkages.
C. Branched polysaccharide - Cellulose is an unbranched polysaccharide
**A. Has beta 1-4 glycosidic links - CORRECT ANSWER – beta 1-4 links in cellulose are not hydrolyzed by the digestive enzymes**
B. A storage polysaccharide found in liver – Glycogen is the storage polysaccharide found in the liver and muscle

4. A 2-year-old boy is brought to the physician by his mother because of a 3-week history of tripping over objects. Physical examination shows bilateral cataracts. Laboratory tests confirm inherited galactokinase deficiency. Examination of the lens shows an abnormal accumulation of a metabolite. Which of the following enzymes is most likely required for the formation of this metabolite in the lens of this patient?

Explanation: The question is testing on accumulation of galactitol (sugar alcohol) in the lens of a child with galactosemia. Galactitol is formed from galactose by the enzyme aldose reductase in the lens.

A. Glucose 6-phosphatase – This is the last enzyme of gluconeogenesis and glycogen breakdown. Deficiency of this enzyme is von Gierke disease.
B. Hexokinase – found in all tissues except liver and beta cells of pancreas. Involved in phosphorylating glucose to glucose 6-phosphate

C. Glucose 6-phosphate dehydrogenase – This is an enzyme of the pentose phosphate pathway; forms NADPH; Deficiency results in hemolytic anemia due to accumulation of reactive oxygen species
**D. Aldose reductase – CORRECT ANSWER – Aldose reductase converts galactose to galactitol in children with untreated galactosemia. Aldose reductase also converts glucose to sorbitol in patients with uncontrolled diabetes mellitus.**
E. Pyruvate dehydrogenase – converts pyruvate to acetyl CoA (mitochondrial enzyme)

5. A 62-year-old man comes to the physician because of a 1-month history of poor vision especially at night. He has a 15-year history of type 2 diabetes mellitus. Physical examination shows bilateral opacity of the lens. Laboratory studies show HbA1c is 9%. Which of the following best represents the chemical nature of the compound most likely responsible for the cataracts in this patient?

A. A sugar acid – Example is uronic acid
B. An amino sugar – Example is galactosamine or glucosamine
C. A homopolysaccharide – Starch, glycogen and cellulose are all polysaccharides made up of glucose
D. A heteropolysaccharide – Glycosaminoglycans or mucopolysaccharides
**E. A sugar alcohol – CORRECT ANSWER – Sorbitol is formed from glucose by the action of aldose reductase**

## Glycolysis and Pentose phosphate pathway and their role in metabolism:

1. A researcher studying metabolism inhibits the enzyme glucose 6-phosphate dehydrogenase in the liver. Which of the following pathways is most likely affected in the liver?
   A. Beta oxidation
   B. Gluconeogenesis
   C. TCA cycle
   D. Cholesterol synthesis
   E. Ketone body synthesis

2. A 42-year-old man is brought to the physician because he has refused to eat and is on a hunger strike for 5 days. Which of the following metabolic changes is most likely observed in the liver on the fifth day?

   A. Intracellular cAMP levels are very low
   B. Pyruvate kinase is dephosphorylated

C. Fructose 2,6 bisphosphate levels are low
D. Phosphofructokinase-2 is active
E. Glucokinase is very active

3. A 50-year-old man comes to the physician for a follow-up examination. He has been advised to undergo the oral glucose tolerance test. He is given 50 grams of glucose orally as part of the oral glucose tolerance test. 30 minutes after ingesting glucose, which of the following enzymes would most likely be most active in the liver of this patient?

   A. Protein kinase A
   B. Glucose 6-phosphatase
   C. Pyruvate carboxylase
   D. Fructose 1,6 bisphosphatase
   E. Phosphofructokinase-1

4. A 45-year-old woman comes to the physician for a follow-up examination. She has no family history of type 2 diabetes. Laboratory studies over the past 6 months show plasma glucose levels are 105mg/dL on more than one occasion. An oral glucose tolerance test is performed. Thirty minutes after ingestion of 50 grams of glucose, which of the following metabolic changes are most likely observed in the liver of this patient?

   A. Increased activity of gluconeogenesis
   B. Phosphorylation of the bifunctional enzyme
   C. Increased levels of intracellular cAMP
   D. Increased activity of phosphofructokinase-2
   E. Decreased activity of pyruvate kinase

5. A 5-year-old girl is brought to the physician by her parents because she has a 2-week history of swelling of her feet. Her pulse is 120/minute. Physical examination shows bilateral pitting pedal edema. Her dietary history shows a diet high in refined foods and avoidance of whole grain products. A diagnosis of thiamine deficiency is made. Which of the following pathways most likely utilizes this vitamin as a coenzyme?
   A. Gluconeogenesis
   B. Glycolysis
   C. Pentose phosphate pathway
   D. Heme degradation
   E. Fatty acid synthesis

6. A researcher is studying gluconeogenesis and glycolysis in hepatocytes. Which of the following enzymes is most likely used only in gluconeogenesis but not in glycolysis?
   A. Pyruvate kinase
   B. Phosphofructokinase-1
   C. Fructose 1,6-bisphosphatase
   D. Aldolase B
   E. Glyceraldehyde 3-P dehydrogenase

Answer Key:

1. D
2. C
3. E
4. D
5. C
6. C

## Short Explanations:

1. A researcher studying metabolism inhibits the enzyme glucose 6-phosphate dehydrogenase in the liver. Which of the following pathways is most likely affected in the liver?
   **Explanation:** G6PD is the first enzyme of the pentose phosphate pathway which is important for the formation of NADPH; NADPH formed in the PPP is used for anabolic pathways like fatty acid biosynthesis, cholesterol synthesis and steroid hormone synthesis. It is also used for regeneration of reduced form of glutathione (GSH) in RBCs.

   A. Beta oxidation – is a catabolic pathway which forms NADH and FADH2 for the ATP formation by the ETC
   B. Gluconeogenesis – does not use NADPH
   C. TCA cycle – Forms NADH which is used for generation of ATP by ETC
   **D. Cholesterol synthesis – CORRECT ANSWER – HMGCoA reductase enzyme uses NADPH**
   E. Ketone body synthesis – uses NADH to form 3-hydroxybutyrate from acetoacetate

2. A 42-year-old man is brought to the physician because he has refused to eat and is on a hunger strike for 5 days. Which of the following metabolic changes is most likely observed in the liver on the fifth day?
   **Explanation:** The man is in starvation; Insulin: glucagon ratio would be low in this individual. Glucagon binds to its receptors on the liver cells and activates adenylate cyclase which increases cAMP; Increased cAMP activates protein kinase A; Key enzymes would be present in the phosphorylated state due to activation of protein kinase A;

   A. Intracellular cAMP levels are very low – The intracellular cAMP levels would be high because of increased glucagon
   B. Pyruvate kinase is dephosphorylated – Pyruvate kinase (regulated enzyme of glycolysis) is phosphorylated by protein kinase A and is inactivated. Remember, pyruvate kinase is dephosphorylated by insulin in the well fed state (high blood glucose level) and is active in the dephosphorylated state
   **C. Fructose 2,6 bisphosphate levels are low – CORRECT ANSWER – The bifunctional enzyme is phosphorylated – This activates FBPase-2 activity which reduces the levels of fructose 2,6 bisphosphate in the liver. Low levels of fructose 2,6 bisphosphate will reduce activity of PFK-1 and thus reduce flux through glycolysis**

D. Phosphofructokinase-2 is active Phosphofructokinase-2 is active – PFK-2 (Bifunctional enzyme which regulates the levels of fructose 2,6-bisphosphate) is phosphorylated and less active. As a result, fructose 2,6-bisphosphate levels are low which decreases the flux through glycolysis. The flux through gluconeogenesis is increased in the fasting state under the influence of glucagon
E. Glucokinase is very active – Glucokinase activity is reduced as the enzyme has a high Km for glucose and low plasma glucose in the fasting state reduces the activity of the enzyme

3. A 50-year-old man comes to the physician for a follow-up examination. He has been advised to undergo the oral glucose tolerance test. He is given 50 grams of glucose orally as part of the oral glucose tolerance test. 30 minutes after ingesting glucose, which of the following enzymes would most likely be most active in the liver of this patient?

   **Explanation:** The man would have increased blood glucose that stimulates the release of insulin and high insulin: glucagon ratio in circulation

A. Protein kinase A – Activated by cAMP which increases in the presence of glucagon
B. Glucose 6-phosphatase – Is an irreversible enzyme of gluconeogenesis which is less active at this time
C. Pyruvate carboxylase Pyruvate carboxylase - Is an irreversible enzyme of gluconeogenesis which is less active when the insulin: glucagon ratio is high
D. Fructose 1,6 bisphosphatase - Is an irreversible enzyme of gluconeogenesis which is less active at this time; It is inhibited due to high levels of fructose 2,6 bisphosphate
**E. Phosphofructokinase-1 – CORRECT ANSWER - Key regulated enzyme of glycolysis which is allosterically activated by high levels of fructose 2,6 Bisphosphate – which is formed by increased PFK-2 activity under the influence of insulin**

4. A 45-year-old woman comes to the physician for a follow-up examination. She has no family history of type 2 diabetes. Laboratory studies over the past 6 months show plasma glucose levels are 105mg/dL on more than one occasion. An oral glucose tolerance test is performed. Thirty minutes after ingestion of 50 grams of glucose, which of the following metabolic changes are most likely observed in the liver of this patient?

**Explanation:** Thirty minutes after an OGTT, plasma glucose levels are increased which stimulate the release of insulin and increase the insulin: glucagon ratio in circulation

A. Increased activity of gluconeogenesis – Gluconeogenesis becomes less active due to inhibition of fructose 1,6-BPase by high levels of fructose 2,6-bisphosphate
B. Phosphorylation of the bifunctional enzyme – Increased insulin:glucagon ratio causes dephosphorylation of the bifunctional enzyme
C. Increased levels of intracellular cAMP – Increased insulin: glucagon ratio decreases the intracellular cAMP levels
**D. Increased activity of phosphofructokinase-2: Correct answer. The bifunctional enzyme is dephosphorylated in the presence on high insulin: glucagon ratio. This activates PFK-2 and reduces the activity of FBPase-2. As a result, the fructose 2,6 bisphosphate levels are**

**increased, which in turn allosterically activates PFK-1 (increases flux through glycolysis). Increased levels of fructose 2,6-bisphosphate inhibits fructose 1,6-bisphosphatase and decreases gluconeogenesis**

E. Decreased activity of pyruvate kinase – Pyruvate kinase is dephosphorylated and is active. Also, high levels of fructose 1,6-bisphosphate acts as a feed-forward activator of pyruvate kinase

5. A 5-year-old girl is brought to the physician by her parents because she has a 2-week history of swelling of her feet. Her pulse is 120/minute. Physical examination shows bilateral pitting pedal edema. Her dietary history shows a diet high in refined foods and avoidance of whole grain products. A diagnosis of thiamine deficiency is made. Which of the following pathways most likely utilizes this vitamin as a coenzyme?
**Explanation:** TPP is the coenzyme form of thiamine (B1). TPP is required for transketolase (pentose phosphate pathway), PDH complex and alpha-ketoglutarate DH complexes.
   A. Gluconeogenesis – requires biotin
   B. Glycolysis – uses NAD (Niacin, B3)
   **C. Pentose phosphate pathway – CORRECT ANSWER – Transketolase requires TPP**
   D. Heme degradation – requires PLP (pyridoxine; B6)
   E. Fatty acid synthesis – requires biotin

6. A researcher is studying gluconeogenesis and glycolysis in hepatocytes. Which of the following enzymes is most likely used only in gluconeogenesis but not in glycolysis?
   **A.** Pyruvate kinase - An irreversible reaction in glycolysis (does not take part in gluconeogenesis)
   B. Phosphofructokinase-1 - An irreversible reaction in glycolysis (does not take part in gluconeogenesis)
   C. **Fructose 1,6-bisphosphatase – CORRECT ANSWER – An irreversible reaction in gluconeogenesis (does not take part in glycolysis)**
   D. Aldolase B - A reversible reaction in glycolysis and gluconeogenesis. Takes part in both glycolysis and gluconeogenesis
   E. Glyceraldehyde 3-P dehydrogenase - A reversible reaction in glycolysis and gluconeogenesis – It forms NADH; Used in both glycolysis and gluconeogenesis

# Practice Questions: Gluconeogenesis

## Gluconeogenesis (MCQ)

1. A 42-year-old man is brought to the emergency department by his wife because of a 2-hour history of abdominal pain, vomiting, and confusion. His wife reports that he has a history of chronic alcohol use, was binge drinking, and has not eaten for the past 2 days. Results of laboratory studies are shown in the table below. After administration of glucagon, the patient's blood glucose increases only slightly, from 42 mg/dL to 50 mg/dL (normal: 70-100 mg/dL). The physician suspects inhibition of a specific pathway responsible for maintaining blood glucose levels, especially after a 2-day fast. Which of

the following best describes an enzyme that catalyzes an irreversible step of this pathway?

| Laboratory test | Patient | Reference |
| --- | --- | --- |
| Lactate | 6.5 mmol/L | 0.5-2.2 mmol/L |
| AST | 85 U/L | 10-40U/L |
| ALT | 50 U/L | 7-56U/L |
| Blood Glucose | 42mg/dL | 70-100mg/dL |

A. Phosphofructokinase-1
B. Fructose 1,6-bisphosphatase
C. Hexokinase
D. Pyruvate kinase
E. Glucokinase

2. A 2-year-old boy is brought to the emergency department by his father because of a 6-hour history of vomiting and a 12-hour history of decreased appetite. The child recently had an upper respiratory tract infection. Physical examination shows hepatomegaly. Laboratory studies show blood glucose is 35 mg/dL (normal: 70-100 mg/dL), plasma ketones are low, and the plasma acylcarnitine profile shows medium-chain acylcarnitines are elevated. The physician explains that the patient is deficient in an enzyme in a pathway that produces acetyl-CoA, an allosteric activator of an enzyme in gluconeogenesis. Which of the following best describes this enzyme in the pathway?
   A. Pyruvate carboxylase
   B. Fructose 1,6-bisphosphate
   C. Phosphoenolpyruvate carboxykinase (PEPCK)
   D. Glucose 6-phosphatase
   E. Malate dehydrogenase

3. While studying gluconeogenesis, a biochemist discovered an enzyme that catalyzes the conversion of fructose 1,6-bisphosphate to fructose-6-phosphate, which is the rate-limiting step of gluconeogenesis. Which of the following is most likely an allosteric inhibitor of this enzyme?

   A. Fructose 2,6-bisphosphate
   B. Acetyl CoA
   C. ATP
   D. Glucose 6-phosphate

E. Citrate

4. An 18-year-old man presents to the emergency department because of dizziness after his first day in the gym, in which he spent 30 minutes using the treadmill at a moderate pace. He has been skipping meals and doing moderate exercise to increase his physical activity and optimize his BMI. Which of the following metabolic changes is most likely occurring in his liver during the fast?

   A. Fatty acid oxidation produces acetyl-CoA, which directly enters gluconeogenesis
   B. Fatty acid oxidation generates acetyl-CoA, which activates gluconeogenesis
   C. Beta-oxidation converts fatty acids into lactate for anaerobic ATP generation
   D. Beta-oxidation provides glucose-6-phosphate for the pentose phosphate pathway
   E. Beta-oxidation is inhibited, and gluconeogenesis is active

5. A 58-year-old woman is brought to the emergency room by her daughter after several days of loss of appetite and poor food intake. She has a 2-year history of type 2 diabetes mellitus and is not adherent to her medications. Physical examination shows pale and dry mucus membranes. Laboratory tests show blood glucose concentration is 375 mg/dL (Normal: 70-100 mg/dL) and an elevated serum free fatty acid concentration. Which of the following is most likely observed in the patient's hepatocytes?

   A. Increased fructose 2-6 bisphosphate concentration activating phosphofructokinase-1
   B. Increased AMP concentration inhibiting fructose-1-6 bisphosphatase
   C. Increased acetyl-CoA concentration, activating pyruvate carboxylase
   D. Increased insulin concentration activates pyruvate kinase
   E. Increased ADP activating phosphoenolpyruvate carboxykinase

**Gluconeogenesis (MCQ with explanations)**

1. A 42-year-old man is brought to the emergency department by his wife because of a 2-hour history of abdominal pain, vomiting, and confusion. His wife reports that he has a history of chronic alcohol use, was binge drinking, and has not eaten for the past 2 days. Results of laboratory studies are shown in the table below. After administration of glucagon, the patient's blood glucose increases only slightly, from 42 mg/dL to 50 mg/dL (normal: 70-100 mg/dL). The physician suspects inhibition of a specific pathway responsible for maintaining blood glucose levels, especially after a 2-day fast. Which of the following best describes an enzyme that catalyzes an irreversible step of this pathway?

| **Laboratory test** | **Patient** | **Reference** |
|---|---|---|
| Lactate | 6.5 mmol/L | 0.5-2.2 mmol/L |
| AST | 85 U/L | 10-40U/L |
| ALT | 50 U/L | 7-56U/L |
| Blood Glucose | 42mg/dL | 70-100mg/dL |

A. Phosphofructokinase-1
B. Fructose 1,6-bisphosphatase
C. Hexokinase
D. Pyruvate kinase
E. Glucokinase

**Correct:** B

**Explanation:**

After glucagon administration, gluconeogenesis should increase blood glucose levels. However, because this patient has been binge drinking for the past 2 days, alcohol metabolism has led to a high $NADH/NAD^+$ ratio. This inhibits the entry of specific intermediates (pyruvate and oxaloacetate) into gluconeogenesis, thereby decreasing gluconeogenesis and causing hypoglycemia.

-Of the enzymes listed, **fructose-1,6-bisphosphatase** is the only one involved in the irreversible steps of gluconeogenesis.

**-Phosphofructokinase-1** is the rate-limiting enzyme of glycolysis and catalyzes an irreversible step in that pathway.

**-Hexokinase** and **pyruvate kinase** are enzymes in glycolysis that catalyze irreversible steps of that pathway.

2. A 2-year-old boy is brought to the emergency department by his father because of a 6-hour history of vomiting and a 12-hour history of decreased appetite. The child recently had an upper respiratory tract infection. Physical examination shows hepatomegaly. Laboratory studies show blood glucose is 35 mg/dL (normal: 70-100 mg/dL), plasma ketones are low, and the plasma acylcarnitine profile shows medium-chain acylcarnitines are elevated. The physician explains that the patient is deficient in an enzyme in a pathway that produces acetyl-CoA, an allosteric activator of an enzyme in gluconeogenesis. Which of the following best describes this enzyme in the pathway?
   A. Pyruvate carboxylase
   B. Fructose 1,6-bisphosphate
   C. Phosphoenolpyruvate carboxykinase (PEPCK)
   D. Glucose 6-phosphatase
   E. Malate dehydrogenase

**Correct:** A

**Explanation:**

A. **Pyruvate carboxylase:** Pyruvate → Oxaloacetate. It is activated by acetyl-CoA, linking fatty acid oxidation to gluconeogenesis. Acetyl CoA is an absolute allosteric activator of pyruvate carboxylase.

B. **Fructose 1,6-bisphosphatase:** Fructose 1,6-bisphosphate → Fructose 6-phosphate, which is the rate-limiting step in gluconeogenesis. This enzyme is allosterically inhibited by fructose 2,6-bisphosphate, not acetyl-CoA.

C. **Phosphoenolpyruvate carboxykinase (PEPCK):** Oxaloacetate → Phosphoenolpyruvate, which is an irreversible step in gluconeogenesis. PEPCK converts oxaloacetate to PEP; not dependent on acetyl-CoA for activation. This enzyme is activated by cortisol.

D. **Glucose 6-phosphatase:** Glucose-6-phosphate → Glucose, which is the final step of gluconeogenesis and glycogenolysis. Glucose-6-phosphatase releases glucose into the bloodstream; not regulated by acetyl-CoA.

E. **Malate dehydrogenase:** Oxaloacetate ↔ Malate (mitochondria ↔ cytosol). This step is important as it allows oxaloacetate to exit mitochondria as malate for gluconeogenesis. This enzyme is also important in the TCA cycle for NADH formation. When the NADH/NAD+ ratio is high, oxaloacetate is trapped as malate. Malate trapping is observed following binge alcohol consumption.

3. While studying gluconeogenesis, a biochemist discovered an enzyme that catalyzes the conversion of fructose 1,6-bisphosphate to fructose-6-phosphate, which is the rate-limiting step of gluconeogenesis. Which of the following is most likely an allosteric inhibitor of this enzyme?

   A. Fructose 2,6-bisphosphate
   B. Acetyl CoA
   C. ATP
   D. Glucose 6-phosphate
   E. Citrate

Correct: A

**Explanation:**

   A. **Fructose 2,6-bisphosphate: Correct: This is a direct allosteric inhibitor of fructose 1,6-bisphosphatase. Produced by the bifunctional enzyme. A high concentration of fructose 2,6-bisphosphate activates glycolysis (PFK-1).**

   B. **Acetyl CoA:** activates pyruvate carboxylase, not fructose 1,6-bisphosphatase.

   C. **ATP:** inhibits Phosphofructokinase-1 in glycolysis, not fructose 1,6-bisphosphatase in gluconeogenesis. AMP activates PFK-1.

   D. **Glucose 6-phosphate:** not an allosteric inhibitor of fructose 1,6-bisphosphatase. Glucose 6-phosphate inhibits hexokinase (product inhibition)

   E. **Citrate:** Citrate inhibits glycolysis (PFK-1). Citrate promotes gluconeogenesis by activating fructose 1,6-bisphosphatase, not inhibiting it

4. An 18-year-old man presents to the emergency department because of dizziness after his first day in the gym, in which he spent 30 minutes using the treadmill at a moderate pace. He has been skipping meals and doing moderate exercise to increase his physical activity and optimize his BMI. Which of the following metabolic changes is most likely occurring in his liver during the fast?

   A. Fatty acid oxidation produces acetyl-CoA, which directly enters gluconeogenesis
   B. Fatty acid oxidation generates acetyl-CoA, which activates gluconeogenesis
   C. Beta-oxidation converts fatty acids into lactate for anaerobic ATP generation
   D. Beta-oxidation provides glucose-6-phosphate for the pentose phosphate pathway
   E. Beta-oxidation is inhibited, and gluconeogenesis is active

**Answer: B**

**Explanation:** β-oxidation of fatty acids generates large amounts of NADH and FADH, which donate electrons to the electron transport chain, driving oxidative phosphorylation and ATP synthesis. This ATP produced during fasting provides the energy to synthesize glucose from smaller molecules during gluconeogenesis. Acetyl-CoA, formed by beta-oxidation, activates gluconeogenesis (absolute activator of pyruvate carboxylase)

A. Fatty acid oxidation produces acetyl-CoA, which directly enters gluconeogenesis: **Incorrect:** β oxidation produces acetyl-CoA; however, it does not enter gluconeogenesis. Instead, acetyl-CoA allosterically activates pyruvate carboxylase, converting pyruvate into oxaloacetate. Note that C-atoms from fatty acids are not direct precursors of glucose during fasting.
B. **Fatty acid oxidation generates acetyl-CoA, which activates gluconeogenesis: Correct: β-oxidation of fatty acids generates large amounts of NADH and FADH, which donate electrons to the electron transport chain, driving oxidative phosphorylation and ATP synthesis. This ATP produced during fasting provides the energy to synthesize glucose from smaller molecules during gluconeogenesis. Acetyl-CoA, formed by beta-oxidation, activates gluconeogenesis (absolute activator of pyruvate carboxylase)**
C. Beta-oxidation converts fatty acids into lactate for anaerobic ATP generation: **Incorrect:** β oxidation metabolizes fatty acids into NADH, FADH, and acetyl CoA. Lactate is produced during anaerobic glycolysis because of the reduction by NADH using lactate dehydrogenase.
D. Beta-oxidation provides glucose-6-phosphate for the pentose phosphate pathway: **Incorrect:** Glucose 6-phosphate is synthesized from the first step of glycolysis by hexokinase/glucokinase. It can be used in the pentose phosphate pathway, continue through glycolysis, or be used in the synthesis of glycogen (in skeletal muscle and the liver).
E. Beta-oxidation is inhibited, and gluconeogenesis is active: **Incorrect:** β oxidation is active in the liver during fasting, and gluconeogenesis is also active.

5. A 58-year-old woman is brought to the emergency room by her daughter after several days of loss of appetite and poor food intake. She has a 2-year history of type 2 diabetes mellitus and is not adherent to her medications. Physical examination shows pale and dry mucus membranes. Laboratory tests show blood glucose concentration is 375 mg/dL (Normal: 70-100 mg/dL) and an elevated serum free fatty acid concentration. Which of the following is most likely observed in the patient's hepatocytes?

A. Increased fructose 2-6 bisphosphate concentration activating phosphofructokinase-1
B. Increased AMP concentration inhibiting fructose-1-6 bisphosphatase
C. Increased acetyl-CoA concentration, activating pyruvate carboxylase
D. Increased insulin concentration activates pyruvate kinase
E. Increased ADP activating phosphoenolpyruvate carboxykinase

**Answer: C**

**Explanation:** In a type 2 diabetic, there is a low insulin: glucagon ratio and insulin resistance. β-oxidation is increased, producing acetyl-CoA, which allosterically activates pyruvate carboxylase, the first committed step in gluconeogenesis. Note that insulin normally inhibits gluconeogenesis. Insulin resistance/ deficiency increases hepatic gluconeogenesis, leading to hyperglycemia.

A. Increased fructose 2-6 bisphosphate concentration activating phosphofructokinase-1: **Incorrect:** Fructose 2-6 bisphosphate is produced by PFK-2 (Bifunctional enzyme). It would inhibit phosphofructokinase-1, not activate it.
B. Increased AMP concentration inhibiting fructose-1-6 bisphosphatase: **Incorrect:** AMP is a marker of low energy status in the muscle cell; it activates PFK-1 and glycolysis.
C. **Increased acetyl-CoA concentration, activating pyruvate carboxylase: Correct**
D. Increased insulin concentration activates pyruvate kinase: **Incorrect:** Insulin is not an allosteric activator of pyruvate kinase; pyruvate kinase is allosterically activated by Fructose 1,6-bisphosphate (feedforward activator). In contrast, it is allosterically inhibited by ATP and alanine.
E. Increased ADP activating phosphoenolpyruvate carboxykinase: **Incorrect:** PEPCK is not allosterically activated by ADP. PEPCK does not have an allosteric regulator, but it is controlled by substrate availability (oxaloacetate), gene transcription by hormones (upregulated by increased glucagon and cortisol).

# Practice questions for PDH, TCA cycle, ETC, and oxidative phosphorylation

1. A researcher is studying mitochondrial oxidative phosphorylation in the laboratory. To an isolated mitochondrial preparation, 2,4-dinitrophenol is added. Which of the following is most likely observed in the mitochondria following addition of this compound?

A. Increased ATP formation from ADP
B. Decreased translocation of ADP into the mitochondrial matrix
C. Inhibition of electron transfer from complex I to complex III
D. Transport of $H^+$ from the inter-membrane space to the mitochondrial matrix
E. Inhibition of electron transport from complex IV to oxygen

2. A 40-year-old man is admitted to the hospital with a 1-week history of progressive lethargy and intermittent loss of memory. He has a 20-year history of alcohol use. His pulse is 105/min and irregular. Laboratory studies show increased levels of serum lactate. Which of the following conditions is most likely?

A. von Gierke disease

B. Beriberi

C. Leigh syndrome

D. Myoclonic epilepsy with Ragged Red Fibers (MERRF)

E. Early onset Alzheimer disease

3. A 40-year-old man is admitted to the hospital with a 1-week history of progressive lethargy and intermittent loss of memory. He has a 20-year history of alcohol use. His pulse is 105/min and irregular. Laboratory studies show increased anion gap and elevated levels of serum lactate. Which of the following is most likely administered to patient?

A. Oligomycin

B. Thiamine supplements

C. Vitamin B12

D. Carnitine supplements

E. Biotin supplements

4. A group of researchers are studying the electron transport chain and the effect of various inhibitors. They identify an inhibitor that prevents the electron transport by inhibiting complex-III. Which of the following inhibitors is most likely being studied by the researchers?

A. Cyanide

B. Antimycin A

C. Sodium azide

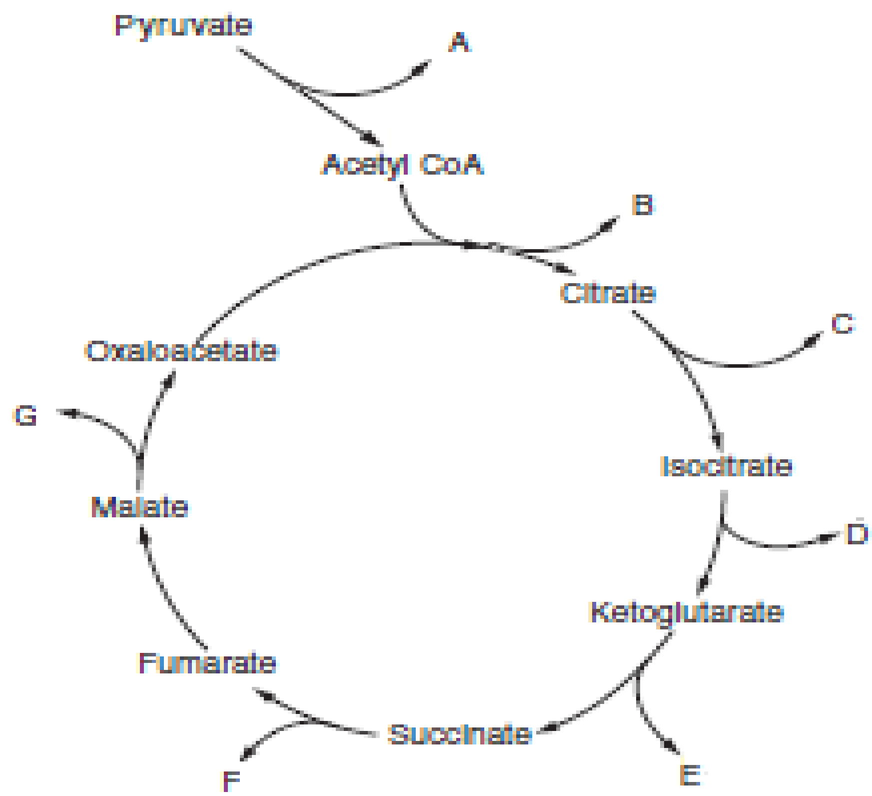
D. Rotenone

E. Amytal

5. A researcher is studying the electron transport chain identify that many small molecules can disrupt ETC and oxidative phosphorylation and act as toxins. She isolates a toxin that acts by preventing the passage of electrons along the electron transport chain. Which of the following compounds is most likely isolated by the researcher?

A. 2,4 dinitrophenol
B. Oligomycin
C. Antimycin A
D. Aspirin
E. Atractyloside

6. A 14-year-old girl is brought to the physician by her mother because of a 2-week history of palpitations and swelling of the feet. She is a picky eater and likes to eat at fast-food restaurants. Her pulse is 120/min, weak and rapid. Physical examination shows bilateral pedal edema. A diagnosis of nutritional thiamine deficiency is made. Which of the following steps of the pathways are most likely to be affected as a result of this deficiency?




A. B and C
B. A and G
C. D and E
D. A and E
E. B and F

7. A researcher is studying the regulation of the pyruvate dehydrogenase complex which links aerobic glycolysis to the TCA cycle. The activity of the enzyme is observed under various metabolic states. Which of the following best describes its regulation?

A. Activation by NADH
B. Activation by acetyl CoA
C. Inhibition by phosphorylation
D. Inhibition by calcium ions
E. Inhibition of PDH kinase by ATP

## Answer key for the practice questions with short explanations.

1. A researcher is studying mitochondrial oxidative phosphorylation in the laboratory. To an isolated mitochondrial preparation, 2,4-dinitrophenol is added. Which of the following is most likely observed in the mitochondria following addition of this compound?

**Comment:** Answer D is correct, 2,4 dinitrophenol is a lipophilic drug that acts as uncoupler of the electron transport/ oxidative phosphorylation. This generates heat instead of ATP. Patients with uncoupler poisoning have hyperthermia

The ETC is not inhibited by 2,4 DNP; rather there is increased electron transport; but reduced ATP formation. The ATP formation from ADP is decreased due to dissipation (loss) of the proton gradient across inner mitochondrial membrane. Electron transport (oxidation) is delinked (uncoupled) from ATP synthesis (phosphorylation).

A. Increased ATP formation from ADP – ATP formation is decreased due to loss of proton gradient
B. Decreased translocation of ADP into the mitochondrial matrix – Occurs in poisoning with Atractyloside and bongkrekic acid. These toxins inhibit the ATP-ADP translocase.
C. Inhibition of electron transfer from complex I to complex III – Amytal and Rotenone are inhibitors of complex-I of the electron transport chain
D. **Transport of $H^+$ from the inter-membrane space to the mitochondrial matrix – Correct answer – The protons that are pumped into the inter membrane space by complexes I, III and IV re-enter the mitochondrial matrix via complex V. The proton gradient is used for ATP synthesis at complex V. In the presence of an uncoupler, the proton gradient is destroyed and the energy is lost as heat. Electron transport occurs but is not linked to ATP synthesis.**
E. Inhibition of electron transport from complex IV to oxygen – Carbon monoxide, cyanide, azide and hydrogen sulfide inhibit electron transport through complex IV. Patients with poisoning by these toxins have severe cellular ATP deficiency and lactic acidosis. Aerobic oxidation is severely limited due to inhibition of the mitochondrial electron transport chain.

2. A 40-year-old man is admitted to the hospital with a 1-week history of progressive lethargy and intermittent loss of memory. He has a 20-year history of alcohol use. His pulse is 105/min and irregular. Laboratory studies show increased levels of serum lactate. Which of the following conditions is most likely?

**Comment:** Patients with alcohol abuse commonly present thiamine deficiency. Decreased activity of the PDH complex results in inhibition of aerobic oxidation and increased conversion of pyruvate to lactate and lactic acidosis.

A. von Gierke disease – Usually presents in infants (1-2 years). Characterized by hypoglycemia, hepatomegaly and lactic acidosis. Deficiency of glucose 6-phosphatase (aka glycogen storage disorder type I)

B. **Beriberi – Correct answer – Patients with alcohol abuse are susceptible to thiamine deficiency and may also present with Wernicke Korsakoff syndrome (nystagmus and speech problems)**

C. Leigh syndrome - Inherited deficiency of complexes of the ETC. Presents in childhood. Patients have developmental delay and lactic acidosis.

D. Myoclonic epilepsy with Ragged Red Fibers (MERRF) – Mitochondrial t-RNA mutation; mainly affects CNS and skeletal muscle.

E. Early onset Alzheimer disease – characterized by accumulation of beta-amyloid precursor protein

3. A 40-year-old man is admitted to the hospital with a 1-week history of progressive lethargy and intermittent loss of memory. He has a 20-year history of alcohol use. His pulse is 105/min and irregular. Laboratory studies show increased anion gap metabolic acidosis and elevated levels of serum lactate. Laboratory studies also show low activity of transketolase. Which of the following is most likely administered to patient?

**Comment:** Patients with alcohol abuse commonly present thiamine deficiency. Decreased activity of the PDH complex results in inhibition of aerobic oxidation and increased conversion of pyruvate to lactate and lactic acidosis. Elevated levels of lactate will increase the anion gap. Lactic acid is a weak acid and gives off protons. The protons are buffered by HCO3 and serum bicarbonate levels fall resulting in high anion gap metabolic acidosis.

A. Oligomycin – is a toxin which inhibits the Complex V (Fo-F1 ATP synthase)

B. **Thiamine supplements – Correct answer** – Thiamine (as TPP) is required for the activity of PDH complex and alpha-ketoglutarate dehydrogenase complex (TCA cycle). TPP is also required by transketolase (Pentose phosphate pathway) and branched chain alpha-keto acid dehydrogenase complex (amino acid metabolism).

C. Vitamin B12 – Deficiency of vitamin B12 presents with neurological and psychiatric symptoms and megaloblastic anemia.

D. Carnitine supplements – are used in patients with fatty acid beta-oxidation defects

E. Biotin supplements – Biotin is a coenzyme required for the activity of carboxylases: Pyruvate carboxylase (gluconeogenesis), acetyl CoA carboxylase (fatty acid synthesis) and propionyl CoA carboxylase.

4. A group of researchers are studying the electron transport chain and the effect of various inhibitors. They identify an inhibitors that prevents the electron transport by inhibiting complex-III. Which of the following inhibitor is most likely being studied by the researchers?

**Comment:** All the listed compounds are inhibitors of electron transport in the electron transport chain (ETC)

A. Cyanide – Inhibits complex IV (AKA cytochrome c oxidase) of ETC. Poisoning results in lactic acidosis and severe reduction in aerobic oxidation. Carbon monoxide (CO) poisoning also inhibits complex IV. CO also competes with oxygen for binding to hemoglobin (Cherry red color of blood; Review from CPR module).

B**. Antimycin A – CORRECT answer - Inhibits complex-III of ETC (aka cyt c reductase). Inhibits electron transport from cyt b to cyt c.**

C. Sodium azide - Inhibit complex IV (AKA cytochrome c oxidase). Inhibit electron transport from cyt $a/a_3$ to molecular oxygen.

D. Rotenone - Rotenone and Amytal inhibit complex I (AKA NADH dehydrogenase). Inhibit electron transfer via FMN to CoQ.

E. Amytal - Rotenone and Amytal inhibit complex I (AKA NADH dehydrogenase). Inhibit electron transfer via FMN to CoQ.

5. A researcher is studying the electron transport chain identify that many small molecules can disrupt ETC and oxidative phosphorylation and act as toxins. She isolates a toxin that acts by preventing the passage of electrons along the electron transport chain. Which of the following compounds is most likely isolated by the researcher?

A. 2,4 dinitrophenol - Uncouples ETC from oxidative phosphorylation by dissipating the proton gradient (uncoupler)
B. Oligomycin - inhibits ATP synthesis at complex V
C. **Antimycin A – Correct -** Antimycin prevents the flow of electrons at complex-III by inhibition of cytochrome reductase
D. Aspirin - Uncouples ETC from oxidative phosphorylation by dissipating the proton gradient (uncoupler)
E. Atractyloside – And Bongkrekic acid inhibits ADP- ATP translocase

6. A 14-year-old girl is brought to the physician by her mother because of a 2-week history of palpitations and swelling of the feet. She is a picky eater and likes to eat at fast-food restaurants. Her pulse is 120/min, weak and rapid. Physical examination shows bilateral pedal edema. A diagnosis of nutritional thiamine deficiency is made. Which of the following steps of the pathways are most likely to be affected as a result of this deficiency?

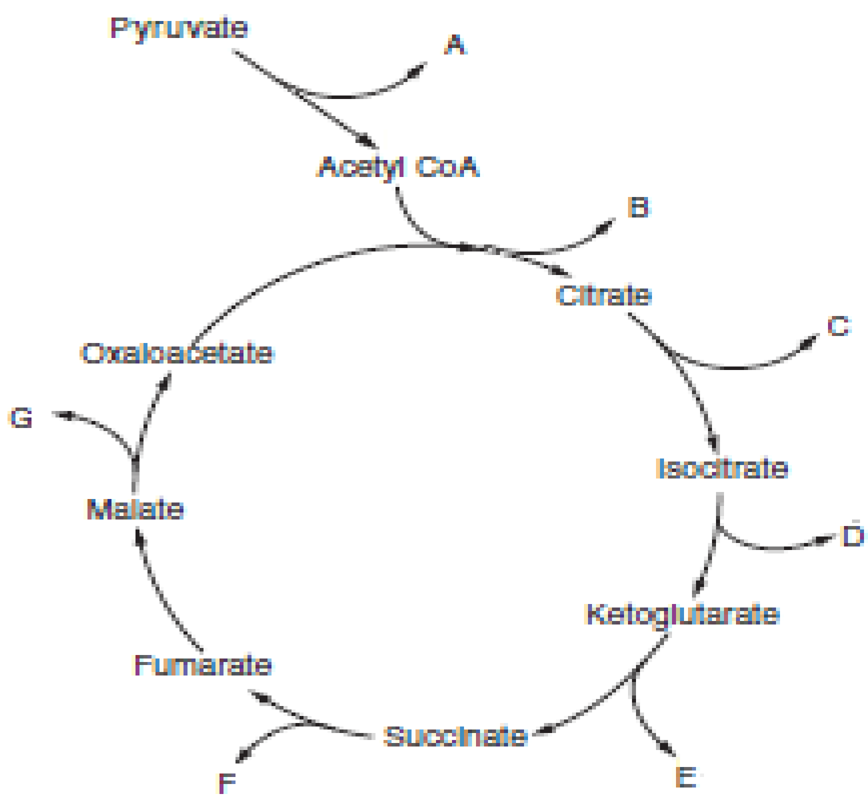
Comment: Pyruvate dehydrogenase (A) and alpha-ketoglutarate dehydrogenase (E) need thiamine-PP as cofactor.

PDH needs TPP but malate dehydrogenase does not need TPP, malate dehydrogenase needs NAD+ in the TCA cycle

Isocitrate dehydrogenase and alpha-ketoglutarate dehydrogenase need both NAD+ but only alpha-ketoglutarate dehydrogenase needs TPP

Is correct

Citrate synthase and succinate dehydrogenase do not need TPP, succinate dehydrogenase needs FAD.




A. B and C - Citrate synthase (B) and aconitase (C) do not need TPP.
B. A and G - PDH (A) needs TPP but malate dehydrogenase (G) does not need TPP, malate dehydrogenase needs NAD+ in the TCA cycle
C. D and E - Isocitrate dehydrogenase (D) and alpha-ketoglutarate dehydrogenase (E) need both NAD+ but only alpha-ketoglutarate dehydrogenase needs TPP

D. **A and E – Correct answer - Pyruvate dehydrogenase and alpha-ketoglutarate dehydrogenase need thiamine-PP as cofactor. Note that both the enzymes need NAD, FAD, CoA, TPP and Lipoic acid as coenzymes. Both are inhibited by Arsenic**
E. B and F - Citrate synthase (B) and succinate dehydrogenase (F) do not need TPP, succinate dehydrogenase needs FAD. Succinate dehydrogenase is inhibited by malonate

7. A researcher is studying the regulation of the pyruvate dehydrogenase complex which links aerobic glycolysis to the TCA cycle. The activity of the enzyme is observed under various metabolic states. Which of the following best describes its regulation?

A. Activation by NADH - it is inhibited by NADH (product inhibition), not activated
B. Activation by acetyl CoA - it is inhibited by acetyl CoA (product inhibition), not activated
C. **Inhibition by phosphorylation – Correct answer: regulation of the PDH complex involves inhibition by phosphorylation. PDH kinase phosphorylates the enzyme and inhibits it.**
D. Inhibition by calcium ions - Calcium ions activate PDH phosphatase and the dephosphorylated PDH is active. In a contracting skeletal muscle increased calcium activates the PDH phosphatase which in turn cause dephosphorylation of PDH complex and activates PDH (facilitates ATP formation which is used for muscle contraction)
E. Inhibition of PDH kinase by ATP - PDH kinase is activated by ATP. High levels of ATP (resting muscle) activates the PDH kinase which phosphorylates PDH and inhibits it.

## Practice MCQ for Glycogen Metabolism and Glycogen storage disorders

1. A 4-month-old boy is brought to the physician by his parents because of a 2-week history of poor feeding. Laboratory studies confirm a diagnosis of Andersen disease. Which of the following clinical features is most likely to indicate this glycogen storage disorder?

A. Cardiomegaly, normal blood glucose

B. Mild hypoglycemia, muscular dystrophy

C. Severe hypoglycemia

D. Muscle cramping, normal blood glucose

E. Hepatomegaly and cirrhosis

2. A research group studies glycogen synthesis in the liver and skeletal muscle of experimental rats. Which of the following best describes this pathway?

A. Glycogen synthase forms alpha 1→ 6 bonds
B. The regulated enzyme is active when phosphorylated
C. Insulin is needed for activation of synthesis in muscle and liver
D. It uses UDP-glucose formed by linkage of UTP to glucose 6-P
E. The regulated enzyme forms the branches in glycogen

3. A research group is comparing glycogen degradation in skeletal muscle and the liver in experimental guinea pigs. Which of the following best describes the pathways in these organs?

A. Activated in the liver by cortisol
B. Regulated by hormonal control in the liver
C. Limit dextrin is an intermediate structure formed by the debranching enzyme
D. Phosphorylase uses the nonreducing ends of glycogen and ATP to form glucose 1-P
E. Glycogen phosphorylase in liver is allosterically activated by AMP

4. A 5-year-old girl is brought to the physician by her parents for a follow-up examination. She has a 6-month history of mild hypoglycemia and muscle weakness. Laboratory studies confirm a diagnosis of a glycogen storage disease which can lead to muscular dystrophy in later life. Which of the following disorders is most likely in this patient?

A. von Gierke disease
B. Pompe disease
C. Cori disease
D. McArdle disease
E. Hers disease

5. A research group is investigating the effect of AMP generated during extensive exercise on skeletal muscle carbohydrate metabolism. It is observed that high levels of AMP allosterically activates an enzyme within skeletal muscle cells. Which of the following enzymes is most likely being studied by the investigators?

A. Isocitrate dehydrogenase

B. Pyruvate kinase

C. Glycogen phosphorylase kinase

D. Phosphofructokinase-1 (PFK-1)

E. Hexokinase

6. An investigator is studying the central role of glucose 6-phosphate in glycogen metabolism and in glycolysis. Which of the following best characterizes the role of this molecule in carbohydrate metabolism?

A. Inhibits glucokinase

B. Used to form glucose 1-phosphate during glycogen synthesis

C. Allosterically inhibits glycogen synthase

D. Formed from glycogen by glycogen phosphorylase

E. Allosterically activates hepatic glycogen phosphorylase

7. A 13-year-old boy is brought to the physician by his mother because of a 2-month history of painful muscle cramping during strenuous exercise. Laboratory studies show no increase in lactate levels following the exercise. A muscle biopsy shows increased amounts of glycogen in his muscle which has a normal structure. Which of the following enzymes is most likely deficient in this patient?

A. Glycogen synthase

B. Pyruvate dehydrogenase

C. Glycogen phosphorylase

D. Glycogen debranching enzyme

E. Phosphofructokinase-1 (PFK-1)

8. A 14-month-old girl is brought to the physician by her mother because she has a 1-month history of twitching movements in the early morning. Laboratory studies show mild hypoglycemia in the fasting state. Which of the following is the most likely diagnosis?

A. von Gierke disease

B. MCAD deficiency

C. Jamaican vomiting sickness

D. Hepatic CPT-I deficiency

E. Hers disease

9. A researcher is studying the central role of the liver in glucose metabolism. In a particular metabolic state, blood glucose concentration is elevated, and hepatocytes exhibit a higher free glucose concentration. Which of the following is most likely observed in hepatocytes under these conditions?

A. Allosteric inhibition of hepatic glycogen phosphorylase

B. Activation of the transport of glycogen into lysosomes for degradation

C. Inhibition of glucokinase release from the nucleus

D. Activation of the glycogen branching enzyme

E. Allosteric inhibition of glycogen synthase

10. A 6-month-old infant is brought to the physician because of a 1-month history of muscle weakness and hypotonia. Physical examination shows hepatomegaly. Imaging studies show massive cardiomegaly. A diagnosis of infantile form of Pompe disease is made. A liver biopsy is performed. Which of the following additional findings is most likely expected in the laboratory data and biopsy results for this patient?

A. Abnormal structure of glycogen

B. Hyperuricemia

C. Normal structure of glycogen in cytosol

D. Moderate hypoglycemia

E. High amount of glycogen in cytosol

Key: 1 **E**, 2 **C**, 3 **B,** 4 **C**, 5 **D**, 6 **B**, 7 **C**, 8 **E,** 9 **A,** 10 **C**.

**Comments and explanations:**

1. A 4-month-old boy is brought to the physician by his parents because of a 2-week history of poor feeding. Laboratory studies confirm a diagnosis of Andersen disease. Which of the following clinical features is most likely to indicate this glycogen storage disorder?

Comment:

A. Cardiomegaly, normal blood glucose – features of Pompe disease

B. Mild hypoglycemia, muscular dystrophy - this is found in Cori disease

C. Severe hypoglycemia - this is found in von Gierke disease

D. Muscle cramping, normal blood glucose - this is found in McArdle disease

E. **Hepatomegaly and cirrhosis – Correct – Inherited deficiency of the branching enzyme. There is accumulation of glycogen with an abnormal structure. Glycogen has long unbranched glucose chains (amylose-like structure) and is attacked by the immune system leading to scarring and familial cirrhosis.**

2. A research group studies glycogen synthesis in the liver and skeletal muscle of experimental rats. Which of the following best describes this pathway?

A. Glycogen synthase forms alpha 1→ 6 bonds - it forms only alpha 1→4 bonds

B. The regulated enzyme is active when phosphorylated - glycogen synthase is inhibited by phosphorylation.

C. **Insulin is needed for activation of synthesis Correct: In both muscle and liver, insulin is needed for activation. Insulin leads to the dephosphorylated glycogen synthase. In the muscle insulin activates GLUT-4 and increases glucose uptake from the blood.**

D. It uses UDP-glucose formed by linkage of UTP to glucose 6-P - UDP-glucose is formed from glucose 1-P and UTP.

E. The regulated enzyme forms the branches in glycogen - Glycogen synthase is the regulated enzyme and does not form the branches. Branching enzyme (not the regulated enzyme) forms the branches (alpha 1→ 6 bonds) in glycogen

3. A research group is comparing glycogen degradation in skeletal muscle and the liver in experimental guinea pigs. Which of the following best describes the pathways in these organs?

A. Activated in the liver by cortisol - Cortisol induces gluconeogenesis by induction of PEP carboxykinase but it does not activate glycogen degradation, this is done by epinephrine and glucagon in the liver.

B. **Regulated by hormonal control in the liver - Correct: The liver glycogen stores are important for glucose release into the blood during fasting, and its degradation is under stringent hormonal control. It occurs only when it is needed for the benefit of the whole body.**

C. Limit dextrin is an intermediate structure formed by the debranching enzyme - limit dextrin is formed by glycogen phosphorylase (when the branch point is about 4-glucose units long). The debranching enzyme uses limit dextrin as a substrate, removes the branch and allows further degradation by glycogen phosphorylase.

D. Phosphorylase uses the nonreducing ends of glycogen and ATP to form glucose 1-P - Phosphorylase performs phosphorolytical cleavage and uses Pi (instead of water and no ATP is used) to form glucose 1-P.

E. Glycogen phosphorylase in liver is allosterically activated by AMP - it is the muscle isozyme of glycogen phosphorylase that is activated by AMP and not liver glycogen phosphorylase.

4. A 5-year-old girl is brought to the physician by her parents for a follow-up examination. She has a 6-month history of mild hypoglycemia and muscle weakness. Laboratory studies confirm a diagnosis of a glycogen storage disease which can lead to muscular dystrophy in later life. Which of the following disorders is most likely in this patient?

A. von Gierke disease – Presents with severe hypoglycemia and glycogen accumulation in liver and kidney. Glucose 6-phosphatase deficiency in liver and kidney. Other features are accumulation of glycogen with normal structure, lactic acidemia, hyperuricemia, hyperlipidemia.

B. Pompe disease – Patients have normal blood glucose levels, muscle weakness and cardiomegaly. Acid maltase (acid glucosidase, alpha 1$\rightarrow$4 glucosidase) deficiency in lysosomes. Characterized by glycogen accumulation in lysosomes (Lysosomal storage disease)

C. **Cori disease - Correct: Cori disease results from deficiency of glycogen debranching enzyme in both liver and muscle. Mild hypoglycemia is found as gluconeogenesis can still be performed.** There is accumulation of glycogen of abnormal structure (Glycogen with short branches; AKA Limit dextrin)

D. McArdle disease – Patients have normal blood glucose levels and muscle weakness during strenuous/ anaerobic activity. Only the muscle isozyme of glycogen phosphorylase is deficient, the hepatic isozyme is normal. Blood lactate levels do not increase when they perform exercise.

E. Hers disease – Patients have mild hypoglycemia but does not lead to muscle weakness. Only the hepatic isozyme of glycogen phosphorylase is deficient, the muscle isozyme is normal.

5. A research group is investigating the effect of AMP generated during extensive exercise on skeletal muscle carbohydrate metabolism. It is observed that high levels of AMP allosterically activates an enzyme within skeletal muscle cells. Which of the following enzymes is most likely being studied by the investigators?

A. Isocitrate dehydrogenase - Enzyme of TCA cycle is allosterically inhibited by NADH and allosterically activated by ADP and calcium ions

B. Pyruvate kinase – Is active in the skeletal muscle at all times in the presence of substrates. Compare to liver pyruvate kinase: The liver enzyme is allosterically activated by F 1,6-bisP formed by PFK-1 in the committed step for glycolysis. The liver uses glucose 6-P for glycogen synthesis, PPP and glycolysis

C. Glycogen phosphorylase kinase - this enzyme is allosterically activated by calcium ions. AMP allosterically activates muscle glycogen phosphorylase.

**D. Phosphofructokinase-1 (PFK-1) – Correct - PFK-1 is allosterically activated by high AMP which is generated in extreme muscle contraction. Glycogen degradation and glycolysis are highly active at high AMP and the pathways are meant to generate ATP for muscle contraction**

E. Hexokinase - this enzyme is only activated by substrate availability inside of the cells.

6. An investigator is studying the central role of glucose 6-phosphate in glycogen metabolism and in glycolysis. Which of the following best characterizes the role of this molecule in carbohydrate metabolism?

A. Inhibits glucokinase - it inhibits hexokinase by product inhibition but not glucokinase.
**B. Used to form glucose 1-phosphate during glycogen synthesis – Correct - Glucose is taken up from the blood, phosphorylated to glucose 6-P and this is changed to glucose 1-P. UDP-glucose is next formed which is a substrate for glycogen synthase.**
C. Allosterically inhibits glycogen synthase - it activates glycogen synthase; In patients with von Gierke disease, high levels of glucose 6-phosphate allosterically activates glycogen synthesis and increases glycogen accumulation in the liver
D. Formed from glycogen by glycogen phosphorylase - forms glucose 1-P and not glucose 6-phosphate
E. Allosterically activates hepatic glycogen phosphorylase - it inhibits phosphorylase

7. A 13-year-old boy is brought to the physician by his mother because of a 2-month history of painful muscle cramping during strenuous exercise. Laboratory studies show no increase in lactate levels following the exercise. A muscle biopsy shows increased amounts of glycogen in his muscle which has a normal structure. Which of the following enzymes is most likely deficient in this patient?

A. Glycogen synthase – Deficiency results in low glycogen levels (not increased glycogen content)

B. Pyruvate dehydrogenase - In PDH deficiency, blood lactate levels would be elevated and may cause lactic acidosis. Aerobic oxidation of pyruvate is reduced due to PDH deficiency. As a result, pyruvate is converted to lactate.

C. **Glycogen phosphorylase – Correct - it is the muscle isozyme that is deficient. Increased glycogen in muscle as it cannot be degraded normally. Patients are unable to perform strenuous activity due to impaired glycogen degradation → less formation of glucose 6-phosphate → less entry into glycolysis → less ATP (causes cramps) and less amounts of lactate formed**

D. Glycogen debranching enzyme – Deficiency results in Cori disease. Accumulation of glycogen with an abnormal structure (glycogen with short branches; limit dextrin)

E. Phosphofructokinase-1 (PFK-1) – Tarui disease; patients have inability to tolerate strenuous activity and hemolysis

8. A 14-month-old girl is brought to the physician by her mother because she has a 1-month history of twitching movements in the early morning. Laboratory studies show mild hypoglycemia in the fasting state. Which of the following is the most likely diagnosis?

A. von Gierke disease – Usually presents with severe hypoglycemia. Fasting should be avoided and nocturnal feeding or using uncooked cornstarch for slow digestion can be used. Patients also have hepatomegaly, nephromegaly, lactic acidosis and hyperuricemia.

B. MCAD deficiency – Presents with severe hypoglycemia and hypoketonemia. The enzyme deficiency allows beta-oxidation of long-chain fatty acyl CoA only to form medium-chain fatty acyl CoAs. This reduces beta-oxidation of fatty acids and the impairs generation of ATP and reduces formation of acetyl CoA. As a result, hepatic gluconeogenesis is strongly reduced. Fasting has to be avoided once a diagnosis is confirmed.

C. Jamaican vomiting sickness - eating unripe ackee fruit leads to inhibition of MCAD and inhibition of beta-oxidation of fatty acids. Presents with severe hypoglycemia

D. Hepatic CPT-I deficiency – Presents with severe hypoglycemia as the carnitine shuttle is deficient and beta-oxidation in mitochondria is severely reduced. Only the hepatic isozyme is deficient, the CPT-I isozyme in muscle is normal.

E. **Hers disease - Correct: the hepatic isozyme of glycogen phosphorylase is deficient leading to hepatomegaly and moderate to mild hypoglycemia.**

9. A researcher is studying the central role of the liver in glucose metabolism. In a particular metabolic state, blood glucose concentration is elevated, and hepatocytes exhibit a higher free glucose concentration. Which of the following is most likely observed in hepatocytes under these conditions?

Explanation: High blood glucose concentration → Increased insulin: glucagon ratio

A. **Allosteric inhibition of hepatic glycogen phosphorylase - Correct: free glucose can accumulate in liver when blood glucose levels are elevated (after a meal). Glucokinase phosphorylates and traps glucose.**

B. Activation of the transport of glycogen into lysosomes for degradation - A small amount of glycogen is transported into lysosomes and degraded by acid maltase; Lysosomal glycogen degradation is not regulated by hormones. The majority of glycogen degradation takes place in the cytosol.

C. Inhibition of glucokinase release from the nucleus - the release of the enzyme back into the cytosol is activated by free glucose; Glucokinase will be more active when there is high hepatocyte glucose concentration.

D. Activation of the glycogen branching enzyme - the branching enzyme is not a regulated enzyme of glycogen synthesis.

E. Allosteric inhibition of glycogen synthase - this enzyme is activated by glucose 6-P but not by free glucose. Glycogen synthase would be active when the blood glucose level is elevated (high insulin).

10. A 6-month-old infant is brought to the physician because of a 1-month history of muscle weakness and hypotonia. Physical examination shows hepatomegaly. Imaging studies show massive cardiomegaly. A diagnosis of infantile form of Pompe disease is made. A liver biopsy is performed. Which of the following additional findings is most likely expected in the laboratory data and biopsy results for this patient?

A. Abnormal structure of glycogen – In Pompe disease, glycogen accumulates (normal structure) in lysosomes

B. Hyperuricemia - this is found in Von Gierke disease due to lactic acidemia. Uric acid and lactate have the same transporter for release into the urine in the kidney. The very high levels of lactate compete and reduce excretion of uric acid in urine.

C. Normal structure of glycogen in cytosol - correct: the cytosolic glycogen structure is normal. Patients with Pompe disease are deficient in lysosomal acid maltase and hence there is accumulation of glycogen in lysosomes in the heart, muscle and liver which can rupture and eventually cause organ damage.

D. Moderate hypoglycemia - the blood glucose level is normal in Pompe disease

E. High amount of glycogen in cytosol – Characterized by glycogen accumulation in lysosomes

## Regulation of carbohydrate metabolism

**Suggestion: Take the 10 MCQs in row and allow yourself 15 min.**

1. A group of investigators is studying metabolic pathways that take place only in the liver and analyzing the synthesis of urea, bile acids, and ketone bodies. Which of the following cells are most likely able to perform gluconeogenesis in addition to the hepatocytes?

A. Subcutaneous fat cells

B. Renal medulla cells

C. Renal cortex cells

D. Skeletal muscle cells

E. Brain stem cells

2. A group of researchers investigate the regulation of glucokinase in the liver. They observe that under a specific metabolic condition, the enzyme is sequestered within the nucleus of hepatocytes. Which of the following metabolic conditions would most facilitate this sequestration?

A. Elevated plasma glucose levels

B. Phosphorylation by protein kinase A

C. Low levels of Glucose 6-phosphate

D. Low plasma glucose levels

E. Elevated levels of Fructose 1,6-bisphosphate

3. A 15-year-old woman is brought to the physician by her parents for a follow-up examination. Laboratory studies over the past 6 months show that her fasting plasma glucose levels were 150 – 160 mg/dL (70-100 mg/dL). Additional laboratory studies confirm the diagnosis of monogenic diabetes (MODY-2). Which of the following enzymes is most likely deficient in this patient?

A. Hexokinase

B. Phosphofructokinase

C. Pyruvate carboxylase

D. Glucokinase

E. Glucose 6-phosphatase

4. Twenty men aged 20-30 years participate in a study of metabolic changes in the fed state. They are given a standard balanced meal and it is observed that their insulin: glucagon ratios are elevated following the meal. It is observed that the hepatic fructose 2,6-bisphosphate levels are elevated. Which of the following enzymes is most likely allosterically activated by this intermediate?

A. Phosphofructokinase -1 (PFK-1)

B. Hexokinase

C. Phosphoenolpyruvate carboxykinase (PEPCK)

D. Pyruvate kinase

E. Glyceraldehyde 3-phosphate dehydrogenase

F. Fructose 1,6-bisphosphatase

5. Twenty women aged 25-30 years participate in a study of metabolism involving the Cori cycle. It is observed that there is transport of metabolites between various organs in this cycle. Which of the following best describes this cycle?

A. Alanine is released by muscle and is used for gluconeogenesis in liver

B. Alanine is released by muscle and is used by heart for energy metabolism

C. Lactate is released by muscle and is used for gluconeogenesis in liver

D. Lactate is released by RBC and is used by the heart for energy metabolism

E. Glutamine is released by muscle and is used for gluconeogenesis in liver

6. A group of researchers observe muscle metabolism in thirty men aged 25 years during a marathon. Their skeletal muscle metabolism and enzymes are studied. Which of the following best describes the regulation of muscle metabolism during the exercise?

A. Glycogen phosphorylase is allosterically activated by calcium ions

B. Hexokinase is activated by AMP

C. PFK-1 is allosterically activated by calcium ions

D. Glycogen phosphorylase kinase is activated by calcium ions

E. Pyruvate kinase is allosterically activated by fructose 2,6 bisphosphate

7. A researcher is observing metabolic changes during strenuous activity in forty men aged 30 years. It is observed that the skeletal muscle cells perform anaerobic glycolysis and use NADH and form NAD+ catalyzed by lactate dehydrogenase. Which of the following enzymes of glycolysis most likely needs NAD+?

A. Aldolase

B. Glyceraldehyde 3-P dehydrogenase

C. Phosphoglycerate kinase

D. Phosphoglycerate mutase

E. Enolase

8. Metabolic changes are observed in a group of healthy women aged 40 years. They are all provided a diet rich in glucose and carbohydrates. It is observed that there is extensive glycolysis in the liver and eventual synthesis of fatty acids and cholesterol. Which of the following best explains the increased flux through glycolysis under these conditions in the healthy liver?

A. Insulin leads to the phosphorylated bifunctional enzyme

B. Pyruvate kinase is feed-forward activated by fructose 1,6-bisP

C. Glucokinase is transported into the nucleus

D. Pyruvate kinase is activated by phosphorylation

E. PFK-1 is activated by ATP

9. A research group investigates the substrates which can be used as precursors of gluconeogenesis in the liver of experimental rats using radiolabeled precursors. The radioactivity in glucose is measured by administration of various precursors. It is observed that one of the radiolabeled precursors when administered to the animals resulted in no appearance of the radiolabel in glucose. Which of the following molecules is most likely responsible for this effect?

A. Alanine

B. Glutamate

C. Leucine

D. Lactate

E. Pyruvate

10. A researcher is studying RBC metabolism and the formation of ATP via glycolysis in these cells. She observes that a reaction results in the substrate level phosphorylation of ADP to ATP

in the observed erythrocytes. Which of the following enzymes is most likely being studied by the researcher?

A. Hexokinase

B. PFK-1

C. Aldolase

D. Phosphoglycerate kinase

E. Enolase

**Answer Key: 1 C, 2 D, 3 D, 4 A, 5 C, 6 D, 7 B, 8 B, 9 C, 10 D.**

**Comments and explanations:**

1. A group of investigators are studying metabolic pathways that take place only in the liver and analyze the synthesis of urea, bile acids and ketone bodies. They observe that gluconeogenesis is performed in the liver and also in another tissue. Which of the following cells are most likely able to perform gluconeogenesis in addition to the hepatocytes?

   A. Subcutaneous fat cells no: All fat cells perform only glycolysis.

   B. Renal medulla cells no: not the medulla cells

   C. **Renal cortex cells - Correct**: **Hepatocytes** and **renal cortex** cells can perform **gluconeogenesis.**

   D. Skeletal muscle cells no: They cannot perform gluconeogenesis.

   E. Brain stem cells no: they cannot perform gluconeogenesis.

2. A group of researchers investigate the regulation of glucokinase in the liver. They observe that under a specific metabolic condition, the enzyme is sequestered within the nucleus of hepatocytes. Which of the following metabolic conditions would most facilitate this sequestration?

A. Elevated plasma glucose levels – This results in the release of the enzyme back into the cytosol

B. Phosphorylation by protein kinase A – Glucokinase is not covalently modified by protein kinase A.

C. Low levels of Glucose 6-phosphate – High levels of Glucose 6-phosphate inhibit hexokinase (product inhibition) and not glucokinase

D. **Low plasma glucose levels – Correct. When the plasma glucose levels are low, glucokinase is sequestered in the nucleus and is less active.**

E. Elevated levels of Fructose 1,6-bisphosphate – Is a feedforward activator of pyruvate kinase in liver

3. A 15-year-old woman is brought to the physician by her parents for a follow-up examination. Laboratory studies over the past 6 months show that her fasting plasma glucose levels were 150 – 160 mg/dL (70-100 mg/dL). Additional laboratory studies confirm the diagnosis of monogenic diabetes (MODY-2). Which of the following enzymes is most likely deficient in this patient?

A. Hexokinase no: It is glucokinase deficiency in liver and pancreatic beta cells.

B. Phosphofructokinase no: Muscle PFK-1 deficiency results in Tarui disease. It is characterized by exercise intolerance and RBC hemolysis

C. Pyruvate carboxylase no: Pyruvate carboxylase deficiency would impair gluconeogenesis and may reduce plasma glucose level

D. **Glucokinase Correct**: **MODY-2 results from glucokinase deficiency**. In the beta cells of pancreas, normal glucokinase activity leads to release of insulin when blood glucose levels are above 5mM/L. In MODY-2, higher plasma glucose levels (about 8 mM/L) is needed for the release of insulin and patients have mild hyperglycemia.

E. Glucose 6-phosphatase no: Deficiency of glucose 6-phosphatase results in fasting hypoglycemia. The disorder is type-I glycogen storage disorder, von Gierke disease

4. Twenty men aged 20-30 years participate in a study of metabolic changes in the fed state. They are given a standard balanced meal and it is observed that their insulin: glucagon ratios are elevated following the meal. It is observed that the hepatic fructose 2,6-bisphosphate levels are elevated. Which of the following enzymes is most likely allosterically activated by this intermediate?

A. **Phosphofructokinase -1 (PFK-1) Correct**: this enzyme is allosterically activated by Fructose 2,6- bisphosphate which overcomes the inhibition by ATP. High AMP levels also allosterically activate PFK-1 (especially important during skeletal muscle contraction)

B. Hexokinase no: this enzyme is not found in remarkable amounts in the liver and is also not activated by Fructose 2,6- bisphosphate. It is activated by free glucose (substrate) and is inhibited by glucose 6-phosphate (product).

C. Phosphoenolpyruvate carboxykinase (PEPCK) no: this is an enzyme of gluconeogenesis and is active during fasting when glucagon is ruling (low insulin: glucagon ratio). The enzyme is also induced by cortisol.

D. Pyruvate kinase no: this is not activated by fructose **2,**6-bisphosphate but by fructose **1,**6 bisphosphate in the liver.

E. Glyceraldehyde 3-phosphate dehydrogenase no: this glycolytic enzyme is important for glycolysis as it uses inorganic phosphate and needs NAD+ but it is not activated by fructose 2,6-bisphosphate.

F. Fructose 1,6-bisphosphatase No: Fructose 2,6-bisphosphate allosterically inhibits fructose 1,6-bisphosphatase (enzyme of gluconeogenesis)

5. Twenty women aged 25-30 years participate in a study of metabolism involving the Cori cycle. It is observed that there is transport of metabolites between various organs in this cycle. Which of the following best describes this cycle?

A. Alanine is released by muscle and is used for gluconeogenesis in liver no: this is described by the alanine-glucose cycle.

B. Alanine is released by muscle and is used by heart for energy metabolism no: the Cori cycle describes hepatic gluconeogenesis using lactate

**C. Lactate is released by muscle and is used for gluconeogenesis in liver Correct:** under conditions of gluconeogenesis, uptake of lactate from muscle leads to release of glucose by the liver.

D. Lactate is released by RBC and is used by the heart for energy metabolism no: lactate is used by the heart for energy metabolism but that is not described in the Cori cycle.

E. Glutamine is released by muscle and is used for gluconeogenesis in liver no: the Cori cycle describes lactate use for hepatic gluconeogenesis

6. A group of researchers observe muscle metabolism in thirty men aged 25 years during a marathon. Their skeletal muscle metabolism and enzymes are studied. Which of the following best describes the regulation of muscle metabolism during the exercise?

A. Glycogen phosphorylase is allosterically activated by calcium ions no: it is activated by AMP. Calcium ions activates glycogen phosphorylase kinase in muscle.

B. Hexokinase is activated by AMP no: AMP activates PFK-1 and activates glycolysis

C. PFK-1 is allosterically activated by calcium ions no: it is allosterically activated by AMP and fructose 2,6-bisP.

D. **Glycogen phosphorylase kinase is activated by calcium ions - Correct answer. Muscle glycogen phosphorylase kinase is activated by calcium ions. Note that during muscle contraction elevated intracellular calcium ion levels will activate, glycogen phosphorylase kinase, PDH complex and TCA cycle to provide ATP.**

E. Pyruvate kinase is allosterically activated by fructose 2,6 bisphosphate - no: the skeletal muscle enzyme is always highly active. Also fructose 2,6 bisphosphate activates PFK-1 and not pyruvate kinase.

7. A researcher is observing metabolic changes during strenuous activity in forty men aged 30 years. It is observed that the skeletal muscle cells perform anaerobic glycolysis and use NADH and form NAD+ catalyzed by lactate dehydrogenase. Which of the following enzymes of glycolysis most likely needs NAD+?

A. Aldolase no: this enzyme cleaves fructose 1,6 bisphosphate to glyceraldehyde 3-phosphate and dihydroxyacetone phosphate and it does not need NAD+

B. **Glyceraldehyde 3 P dehydrogenase Correct**: this enzyme is a dehydrogenase, it uses NAD+ and in this case also inorganic phosphate and forms the energy-rich molecule 1,3 bisphosphoglycerate and NADH. Note: ***Arsenate*** competes with Pi and bypasses the formation of 1,3 BPG.

C. Phosphoglycerate kinase no: this enzyme catalyzes a substrate level phosphorylation reaction and forms ATP in glycolysis. Note: it catalyzes a reversible step of glycolysis and is named in the reverse direction.

D. Phosphoglycerate mutase no: this enzyme shifts the phosphate from 3-phosphoglycerate to 2-phosphoglycerate

E. Enolase no: this enzyme uses 2-phosphoglycerate and forms PEP. Note: ***Fluoride*** inhibits this enzyme.

8. Metabolic changes are observed in a group of healthy women aged 40 years. They are all provided a diet rich in glucose and carbohydrates. It is observed that there is extensive glycolysis in the liver and eventual synthesis of fatty acids and cholesterol. Which of the following best explains the increased flux through glycolysis under these conditions in the healthy liver?

A. Insulin leads to the phosphorylated bifunctional enzyme No: insulin leads to the dephosphorylation of regulated enzymes. It is glucagon that leads to the phosphorylated enzyme using cAMP-messenger and protein kinase A

B. Pyruvate kinase is feed-forward activated by fructose 1,6-bisP - **Correct:** Once F-1,6,bisP is formed by PFK-1, then some of these molecules perform the glycolysis pathway and some molecules allosterically activate pyruvate kinase in the liver.

C. Glucokinase is transported into the nucleus No: Glucokinase moves into the cytosol and becomes active after a meal when the plasma glucose levels are elevated.

D. Pyruvate kinase is activated by phosphorylation no: insulin is ruling and hepatic pyruvate kinase is dephosphorylated. Phosphorylation by PKA leads to inhibition of pyruvate kinase. Glycolysis is inhibited and PEP is used for gluconeogenesis.

E. PFK-1 is activated by ATP no: ATP inhibits PFK-1 in all cells. This inhibition can be overcome by AMP or fructose 2,6-bisphosphate

9. A research group investigates the substrates which can be used as precursors of gluconeogenesis in the liver of experimental rats using radiolabeled precursors. The radioactivity in glucose is measured by administration of various precursors. It is observed that one of the radiolabeled precursors when administered to the animals resulted in no appearance of the radiolabel in glucose. Which of the following molecules is most likely responsible for this effect?

**Explanation:** All glucogenic amino acids (like Alanine and glutamate) can be used as precursors of gluconeogenesis. Lactate and glycerol can also be used for gluconeogenesis. Ketogenic amino acids like leucine and lysine cannot be used for gluconeogenesis. Also, the ketone bodies and acetyl CoA cannot be used for glucose formation.

A. Alanine no: alanine is used to form pyruvate for gluconeogenesis

B. Glutamate no: glutamate is used to form alpha-ketoglutarate for gluconeogenesis

**C. Leucine correct: Leucine and Lysine are purely ketogenic amino acids and cannot be used for gluconeogenesis**

D. Lactate no: lactate is used to form pyruvate for gluconeogenesis

E. Pyruvate no: pyruvate is carboxylated to oxaloacetate for gluconeogenesis

10. A researcher is studying RBC metabolism and the formation of ATP via glycolysis in these cells. She observes that a reaction results in the substrate level phosphorylation of ADP to ATP in the observed erythrocytes. Which of the following enzymes is most likely being studied by the researcher?

A. Hexokinase no this enzyme uses ATP

B. PFK-1 no: this enzyme uses ATP

C. Aldolase no this enzyme cleaves fructose 1,6-bisP

D. **Phosphoglycerate kinase Correct:** this enzyme and pyruvate kinase catalyze substrate level phosphorylation reactions in glycolysis

Please note, that in RBC some of the energy-rich 1,3-bisphosphoglycerate is used to form 2,3-BPG. A deficiency of pyruvate kinase in RBC leads to hemolysis due to lack of ATP.

E. Enolase no: this enzyme uses 2-phosphoglycerate and forms PEP

## Regulation of carbohydrate metabolism:

1. A group of thirty healthy 45 to 50 year-old women are recruited as volunteers for a study of metabolic changes in the fed state and during fasting. They are given a meal containing 50% of calories derived from carbohydrate. Which of the following metabolic change is most likely observed in the liver 30 minutes after the meal?

A. PFK-2 component of the bifunctional enzyme is phosphorylated

B. Levels of fructose 2,6 bisphosphate are low

C. FBP-2 activity of bifunctional enzyme is active

D. Pyruvate kinase is phosphorylated

E. Glycogen synthase is dephosphorylated and active

2. A 38-year-old woman is brought to the emergency department by her husband because of a 6-hour history of weakness. She is on a 4-day voluntary fast. Laboratory studies show her plasma glucose levels are 65 mg/dL (Normal: 70-100 mg/dL). Which of the following statements best describes her metabolism on admission?

A. Blood glucose level is maintained by glycogenolysis

B. Amino acids are the precursors of gluconeogenesis

C. The insulin glucagon ratio is high

D. The key regulated enzymes are in the dephosphorylated state

E. There is increased formation of fructose 2,6 bisphosphate in liver

3. A group of 8-11 year-old boys are recruited for a study of metabolism during stress. They are all exposed to the same stressful conditions and metabolism is observed. Which of the following best describes their metabolism during the stress?

A. Muscle glycogenolysis is inactive

B. Liver glycogen synthase is phosphorylated and active

C. Liver glycogen phosphorylase is phosphorylated and active

D. Liver fructose 2,6-bisphosphate levels are elevated

E. Liver pyruvate kinase is active due to high levels of fructose 1,6 bisphosphate

4. A 25-year-old woman is brought to the emergency department by her friend because of sudden onset of weakness. She was on a 4-day voluntary fast to lose weight. Laboratory studies show plasma glucose level is 70 mg/dL (Normal: 70-100 mg/dL). Which of the following best describes her metabolism on admission?

A. Liver glycogen stores are depleted

B. Muscle glycogenolysis maintains the blood glucose level

C. Glucagon induces glucokinase and PFK-1

D. Insulin induces pyruvate carboxylase and PEPCK

E. Liver cAMP levels are low

5. A 25-year-old woman is brought to the emergency department by her friend because of a sudden onset of dizziness. She is in an intoxicated state. Her friend says that she had large amounts of ethanol in the previous 4 hours. Blood alcohol levels are elevated. Which of the following metabolic change is most likely observed in the liver at the time of admission?

A. Liver NADH/ NAD+ ratio is low

B. Oxaloacetate is converted to malate

C. Liver gluconeogenesis is active

D. Lactate is converted to pyruvate for gluconeogenesis

E. Trapping of inorganic phosphate causes hypoglycemia

Answer Key:

1. E

2. B

3. C

4. A

5. B

# Short explanations: Regulation of carbohydrate metabolism:

1. A group of thirty healthy 45 to 50 year-old women are recruited as volunteers for a study of metabolic changes in the fed state and during fasting. They are given a meal containing 50% of calories derived from carbohydrate. Which of the following metabolic change is most likely observed in the liver 30 minutes after the meal?

**Comment:** At this time, the blood glucose level is elevated and insulin glucagon ratio is high. Key enzymes are in the dephosphorylated state.

A. PFK-2 component of the bifunctional enzyme is phosphorylated – This happens under the influence of glucagon (fasting)

B. Levels of fructose 2,6 bisphosphate are low – This is observed in the fasting state when glucagon levels are elevated; In the fed state, fructose 2,6-bisphosphate levels are elevated.

C. FBP-2 activity of bifunctional enzyme is active – PFK-2 activity of the bifunctional enzyme is active in the dephosphorylated state and FBP-2 is less active

D. Pyruvate kinase is phosphorylated – Pyruvate kinase is dephosphorylated and active; Liver glycolysis is increased

E. Glycogen synthase is dephosphorylated and active **– CORRECT ANSWER – Glycogen synthase is dephosphorylated and is active in the dephosphorylated state. Glucose from the blood is stored as glycogen in the liver. Note that high insulin : glucagon ratio also stimulates glycogen synthesis in muscle.**

2. A 38-year-old woman is brought to the emergency department by her husband because of a 6-hour history of weakness. She is on a 4-day voluntary fast. Laboratory studies show her plasma glucose levels are 65 mg/dL (Normal: 70-100 mg/dL). Which of the following statements best describes her metabolism on admission?

A. Blood glucose level is maintained by glycogenolysis – Liver glycogen stores are depleted after 1 day (17-24 hours) of fasting

B. Amino acids are the precursors of gluconeogenesis **– CORRECT ANSWER – Amino acids derived from the muscle proteolysis are the precursors of gluconeogenesis**

C. The insulin glucagon ratio is high – the insulin glucagon ratio is low

D. The key regulated enzymes are in the dephosphorylated state – The key regulated enzymes are in the phosphorylated state

E. There is increased formation of fructose 2,6 bisphosphate in liver – Fructose 2,6 bisphosphate levels in the liver are low in the fasting state and PFK-1 (glycolysis) is inhibited; Liver gluconeogenesis is active due to activation of fructose 1,6-bisphosphatase

3. A group of 8-11 year-old boys are recruited for a study of metabolism during stress. They are all exposed to the same stressful conditions and metabolism is observed. Which of the following best describes their metabolism during the stress?

A. Muscle glycogenolysis is inactive – Epinephrine activates muscle glycogen breakdown to provide energy for muscle contraction (Fight or flight response)

B. Liver glycogen synthase is phosphorylated and active – Glycogen synthase is less active when it is phosphorylated under the influence of epinephrine; Glycogen synthase in muscle is active in the fed state under the influence of insulin

C. Liver glycogen phosphorylase is phosphorylated and active **– CORRECT ANSWER – Epinephrine increases cAMP in the liver and activates protein kinase A which covalently modifies glycogen phosphorylase kinase which phosphorylates and activates glycogen phosphorylase. This results in breakdown of glycogen to glucose and blood glucose level is elevated**

D. Liver fructose 2,6-bisphosphate levels are elevated – Levels of fructose 2,6 bisphosphate are low and glycolysis is inhibited

E. Liver pyruvate kinase is active due to high levels of fructose 1,6 bisphosphate – Liver glycolysis is less active due to low levels of Fructose 2,6 bisphosphate. As a result, fructose 1,6 bisphosphate levels are low – which reduces activity of pyruvate kinase (Remember, fructose 1,6 bisphosphate is a feedforward activator of pyruvate kinase)

4. A 25-year-old woman is brought to the emergency department by her friend because of sudden onset of weakness. She was on a 4-day voluntary fast to lose weight. Laboratory studies show plasma glucose level is 70 mg/dL (Normal: 70-100 mg/dL). Which of the following best describes her metabolism on admission?

A. Liver glycogen stores are depleted **– CORRECT ANSWER – liver glycogenolysis is inactive at this time as glycogen stores are depleted by the end of first day (17-24 hours) of fasting**

B. Muscle glycogenolysis maintains the blood glucose level – Muscle does not contain glucose 6-phosphatase and cannot increase the blood glucose level during fasting. Muscle utilizes its glycogen for energy during muscle contraction

C. Glucagon induces glucokinase and PFK-1 – Glucagon and fasting induce the enzymes of gluconeogenesis and represses enzymes of glycolysis. Both the listed enzymes are glycolytic enzymes

D. Insulin induces pyruvate carboxylase and PEPCK – Insulin and high carbohydrate diet induce glycolytic enzymes in the liver and repress the gluconeogenic enzymes. Both the listed enzymes are gluconeogenic enzymes

E. Liver cAMP levels are low – Liver cAMP levels are high due to the low insulin glucagon ratio

5. A 25-year-old woman is brought to the emergency department by her friend because of a sudden onset of dizziness. She is in an intoxicated state. Her friend says that she had large amounts of ethanol in the previous 4 hours. Blood alcohol levels are elevated. Which of the following metabolic change is most likely observed in the liver at the time of admission?

**Comment:** Alcohol metabolism forms NADH and increases the ratio of NADH to NAD. This results in trapping of lactate (less formation of pyruvate from lactate). Also, oxaloacetate is converted to malate. Precursors of gluconeogenesis are shunted away from gluconeogenesis resulting in hypoglycemia following binge alcohol consumption.

A. Liver NADH/ NAD+ ratio is low – This ratio is high following ethanol metabolism

B. Oxaloacetate is converted to malate **– CORRECT ANSWER – as a result oxaloacetate is diverted away from gluconeogenesis resulting in hypoglycemia**

C. Liver gluconeogenesis is active - Diversion of pyruvate and oxaloacetate from gluconeogenesis results in severe hypoglycemia. Gluconeogenesis is less active as a result of this diversion of intermediates of gluconeogenesis

D. Lactate is converted to pyruvate for gluconeogenesis – The pyruvate is converted to lactate when NADH/ NAD+ ratio is high. Patients may have **lactic acidosis** and the diversion of pyruvate from gluconeogenesis results in hypoglycemia (Cori cycle is disrupted)

E. Trapping of inorganic phosphate causes hypoglycemia – This is true in patients with hypoglycemia due to hereditary fructose intolerance or classical galactosemia